Huaiyu Jia

Ph.D. Student in FinTech | AI for Blockchain · Prediction Markets · AI Agents

hjia351@connect.hkust-gz.edu.cn | jhuaiyu3@gmail.com | x.com/S33Luke

virusluke3.github.io | Google Scholar | github.com/virusLuke3

Research Interests

Prediction Markets, AI for Blockchain, Market Microstructure, On-chain Analytics, AI Agents

Education

The Hong Kong University of Science and Technology (Guangzhou) **Sep 2024 – Present**
Ph.D. in FinTech, FinTech Thrust Guangzhou, China
Advisor: Dr. Shuo Sun.

University of Electronic Science and Technology of China **Sep 2020 – Jun 2024**
B.Eng., School of Information and Software Engineering Chengdu, China
Undergraduate research with Prof. Dajiang Chen. **GPA**: 3.81/4.0

Selected Publications

Unlocking the Forecasting Economy: A Suite of Datasets for the Full Lifecycle of Prediction Market: [Experiments & Analysis] Preprint, 2026
Huaiyu Jia*, Luofeng Zhou*, Wentao Zhang, Lin William Cong, Siguang Li, Shuo Sun **Equal contribution*

AlphaOpsBench: Benchmarking End-to-End Alpha Strategy Operationalization in Prediction Markets Preprint, 2026
Huaiyu Jia, Mingxuan Zhao, Jincheng Gao, Zifan Peng, Wentao Zhang, Siguang Li, Shuo Sun

Beyond Profit Rankings: Identifying, Understanding, and Evaluating Smart Money in Prediction Markets Preprint, 2026
Huaiyu Jia, Shuo Sun

A Multi-Asset Dataset and Tail-Risk Framework for DeFi Yield Trading Derivatives Preprint, 2026
Huaiyu Jia, Jieshun You, Jingyu Liu, Yizhi Luo, Shuo Sun

A Cross-Protocol Dataset of Blockchain Oracle Incentives and Accountability Preprint, 2026
Huaiyu Jia, Siguang Li, Lin William Cong, Luyao Zhang

Towards Event-Aware Forecasting in DeFi: Insights from On-chain Automated Market Maker Protocols KDD 2026
Huaiyu Jia, Jiehshun You, Yizhi Luo, Jingyu Liu, Shuo Sun

AlphaForgeBench: Benchmarking End-to-End Trading Strategy Design with Large Language Models KDD 2026
Wentao Zhang, Mingxuan Zhao, Jincheng Gao, Jieshun You, **Huaiyu Jia**, Yilei Zhao, Bo An, Shuo Sun

TxSum: User-Centered Ethereum Transaction Understanding with Micro-Level Semantic Grounding EMNLP 2026
Zifan Peng, Jingyi Zheng, Yule Liu, **Huaiyu Jia**, Qiming Ye, Jingyu Liu, Xufeng Yang, Mingchen Li, Qingyuan Gong, Xuechao Wang, Xinlei He

Protected Exercise for Bridgeless Cross-Chain Options Without Upfront Holder Collateral IEEE TDSC 2026
Zifan Peng, Jingyu Liu, **Huaiyu Jia**, Junhao Yang, Yingjie Xue, Xuechao Wang

From Anonymous Wallets to Behavioral Agents: A Calibrated Policy Test Bed for DAO Governance
Peizhe Li, **Huaiyu Jia**, Liang Zhang, Shuo Sun

FS2M: Fuzzy Smart IoT Device Pairing Protocol via Speak to Microphone IJWIS 2025
Huaiyu Jia, Dajiang Chen, Zhidong Xie, Zhiguang Qin

Projects

Polymarket Research Stack

- **Data Engine**. Reconstructs the lifecycle from market creation and fills through oracle resolution.
- **Execution Simulator**. Simulates execution, liquidity, order-book dynamics, and paper-to-live fidelity.
- **Replay Lab**. Replays fill-level, trade-only, and L2 history with oracle-aware settlement.

Polymarket Smart Money

- Evidence-based research on Polymarket wallets, from historical qualification and confirmed trades to verified market evidence.

PolyMonitor | Project · Code

- Public interface for Polymarket prices, fills, oracle events, and lifecycle-aligned market data.

Technical Skills

Programming: Python, Solidity, Rust, C/C++
Blockchain & Tools: Foundry, Hardhat, Anchor, Git, Docker